

INVENTION DISCLOSURE FORM

INNOTECH PROJECT ID

DEPARTMENT - INFORMATION TECHNOLOGY

Ques.1. Title of your invention

ASKLMS – A Unified Learning & Institute Management Platform

Ques.2. Type of Invention

Software System / Web-Based Application (Educational Management System)

Ques.3. Brief Description of your invention.....

ASKLMS is a Unified Learning & Institute Management Platform designed to digitalize the operations of offline small coaching institutes. It integrates course management, student onboarding, fee handling, and content delivery into a single platform. The system provides distinct portals for students and administrators, offering transparency, scalability, and efficient workflow automation. It is currently deployed for Nitant Academy, Saharanpur, serving as a real-world model for similar institutions.

Ques.4. Objective of your invention.....

- To digitalize fee collection and reduce dependency on manual record-keeping.
- To ensure transparency and accountability through role-based access control.
- To make notes, lectures, and learning content easily accessible to students.
- To provide a scalable, mobile-friendly Progressive Web Application (PWA).
- To deploy the system using Docker containers and cloud infrastructure for scalability.
- To provide a centralized platform for managing batches, students, employees, and courses.

Ques.5. How to use the invention

- Admins/Employees log in to manage courses, batches, notes, and fee records.
- **Students** log in to view their enrolled courses, pay fees online, and download notes or recorded lectures.
- The **Homepage** provides institute details, announcements, and registration links.
- The system automates workflows such as payment tracking, student enrollment, and resource sharing through dashboards and role-based modules.

Ques.6. Problem your invention is solving

ASKLMS solves the inefficiencies of traditional coaching management systems, which rely on manual registers, paper-based fee tracking, and limited communication. It addresses:

- Lack of centralized data management.
- Time-consuming manual fee collection.
- Poor transparency between students and management.
- Difficulty in tracking academic progress and records.
- Limited access to learning materials for students.

Ques.7. Purpose and object of Invention

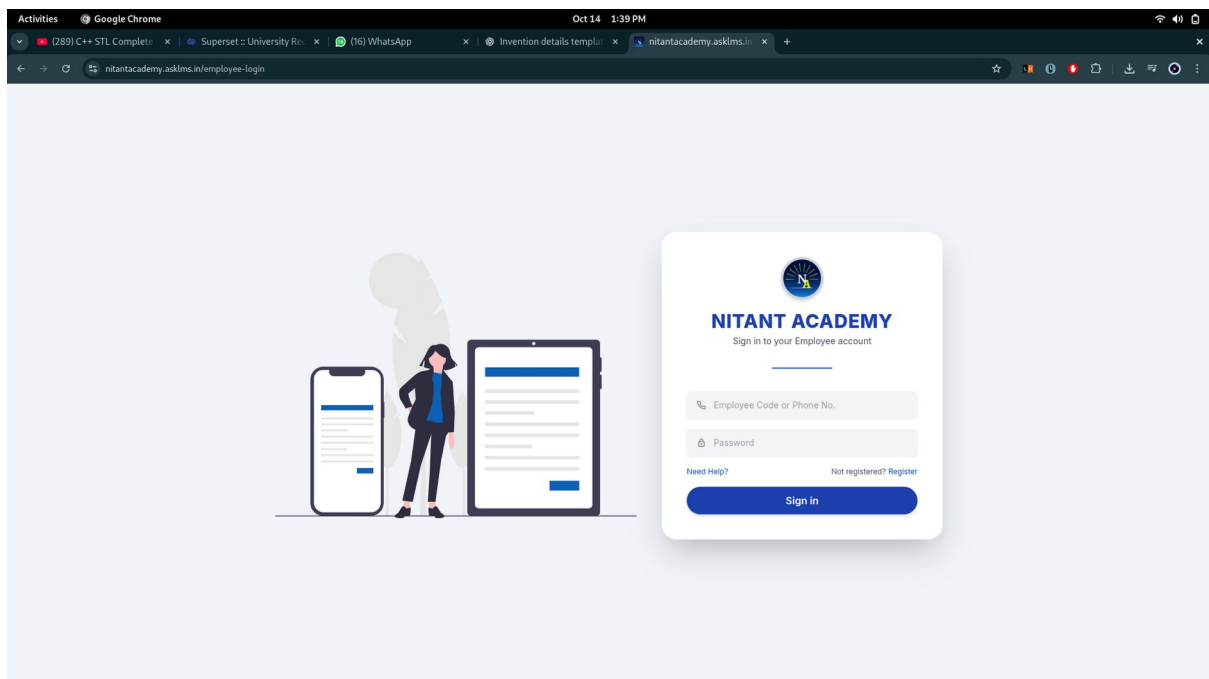
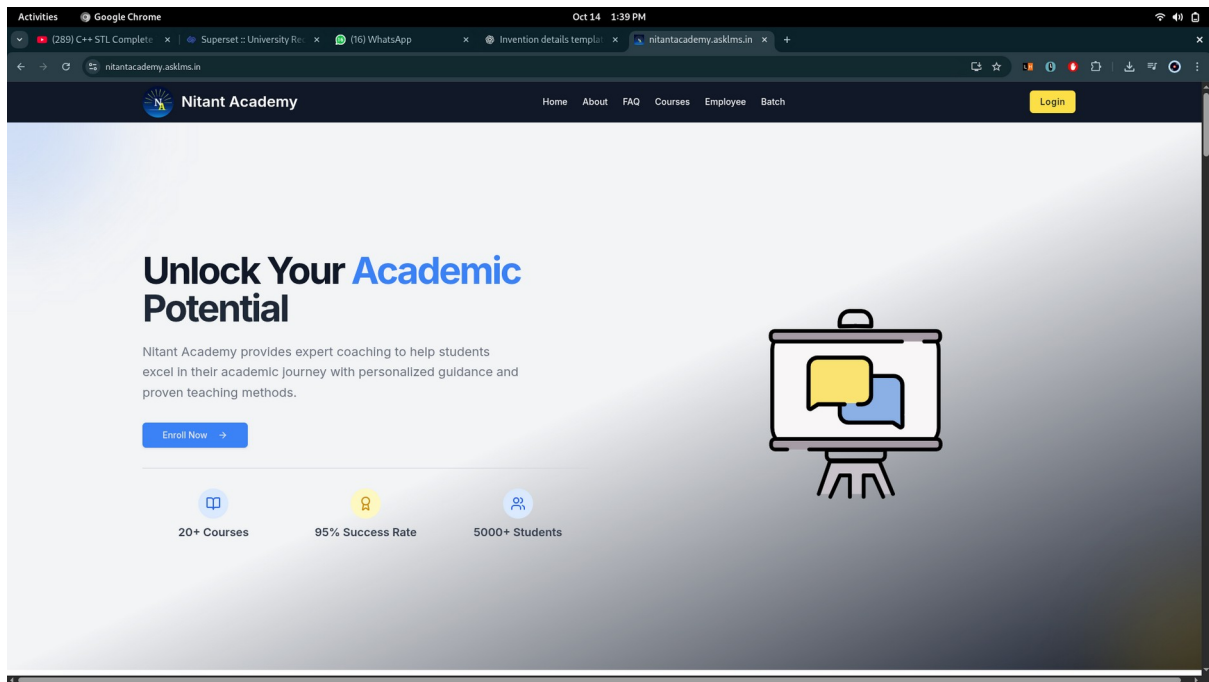
The purpose of ASKLMS is to bridge the gap between traditional offline coaching management and modern digital learning platforms. It simplifies administrative processes, enhances student accessibility, and ensures secure online payment handling. The system aims to make offline coaching institutes digitally capable, efficient, and transparent.

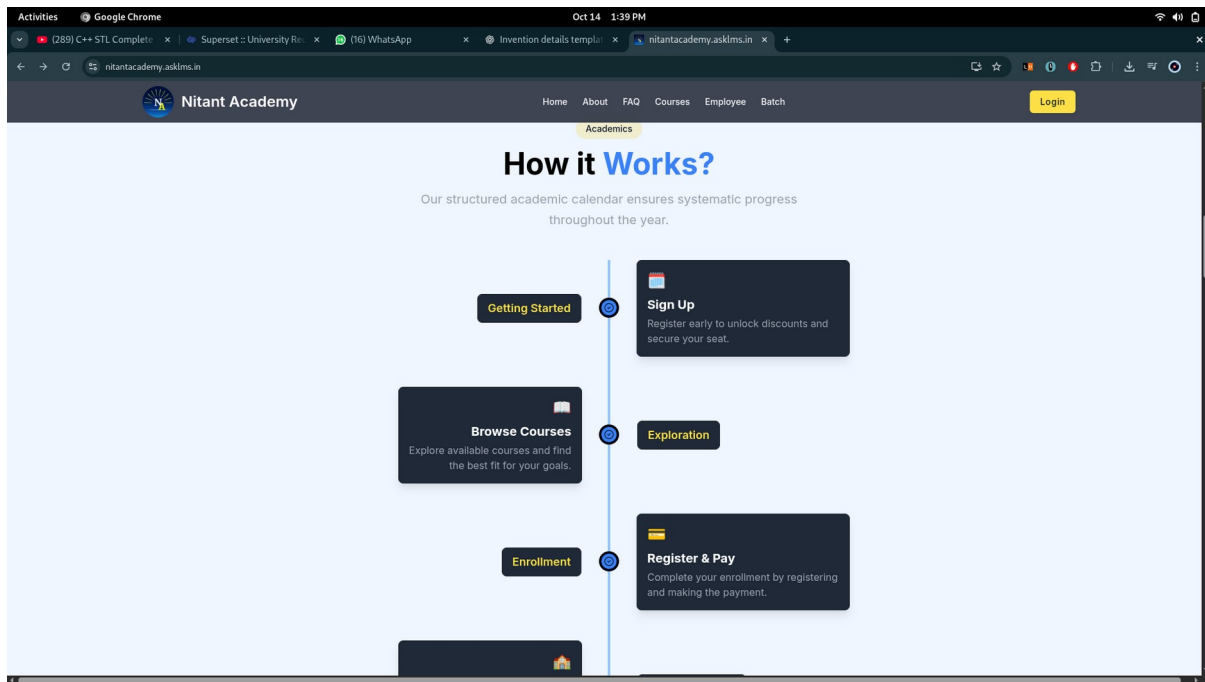
Ques.8. Discuss potential commercial application of the invention.

ASKLMS can be commercialized as a **SaaS (Software as a Service)** or **custom-deployed** solution for:

- Private coaching institutes and tuition centers.
- Test preparation academies (NEET, JEE, UPSC, etc.).
- Schools and educational franchises managing multiple branches.
Institutes can subscribe to ASKLMS with domain-based branding (e.g., *academyname.asklms.in*) for complete digital management of their operations.

Ques.9. Provide any additional material (such as photographs, reports, publications, and references to texts or other information material) which may be helpful to an understanding of the invention identify and indicate the specific relevance of each.





Ques.10. Abstract

ASKLMS is an offline batch management system developed to streamline the operations of coaching institutes. It integrates course, student, and fee management with resource sharing on a unified platform. The system features dedicated portals for students and employees, ensuring transparency, efficiency, and scalability. Currently deployed for Nitant Academy, ASKLMS demonstrates a deployable, institution-ready solution for digital transformation in the education sector.

Ques.11. Summary of the invention

ASKLMS unifies various administrative and learning processes into one software platform. It supports:

- Role-based dashboards for students and employees.
- Secure online fee transactions.
- Course and batch management.
- Digital content and recorded lecture distribution.

Built using **React (frontend)** and **Django (backend)** with **MySQL** as the database, it ensures reliability, scalability, and cross-platform accessibility through a PWA architecture. The

system's deployment using Docker enhances modularity and ease of scaling for multiple institutions.

Ques.12. Detail description of invention with methodology

Architecture Overview:

- **Frontend:** Next.js with Tailwind CSS for responsive UI and PWA support.
- **Backend:** Django for API development.
- **Database:** MySQL for secure and structured data storage.
- **Authentication:** Google OAuth for secure user verification.
- **Deployment:** Hostinger cloud using Cloudflare DNS.

Modules:

1. **Homepage:** Displays institute information, announcements, and quick login/registration links.
2. **Employee/Admin Panel:**
 - Course and category management.
 - Fee and batch registration approvals.
 - Upload and management of notes, lectures, and reports.
 - Role-based access and employee permissions.
3. **Student Portal:**
 - Dashboard with courses and fee summary.
 - Secure fee payment via integrated payment gateway.
 - Download notes and replay recorded sessions.
 - Notifications for updates and achievements.
4. **Reporting Module:** Generates student performance, payment, and activity reports.

Methodology:

The system follows the **MVC (Model-View-Controller)** pattern for modular design. APIs handle communication between the frontend and backend. The application uses secure authentication, error handling, and database optimization for reliability.

Ques.13. Applicant and inventor details:

Field	Details
Applicant / Inventor 1	<i>Srijan Omar</i>
Roll Number	2200290130165
Year	4th Year
Branch	Information Technology
Section	C
Applicant / Inventor 2	<i>Ananya Jain</i>
Roll Number	2200290100020
Year	4th Year
Branch	Computer Science and Engineering
Section	A
Guide Name	Ms. Anupriya Pal
Guide name	Mr. Vipin Deval
Institution	KIET Group of Institutions, Ghaziabad, Uttar Pradesh
Affiliation	Dr. A.P.J. Abdul Kalam Technical University, Lucknow
Outcome	Application Project (Deployable)
Session	2025-26